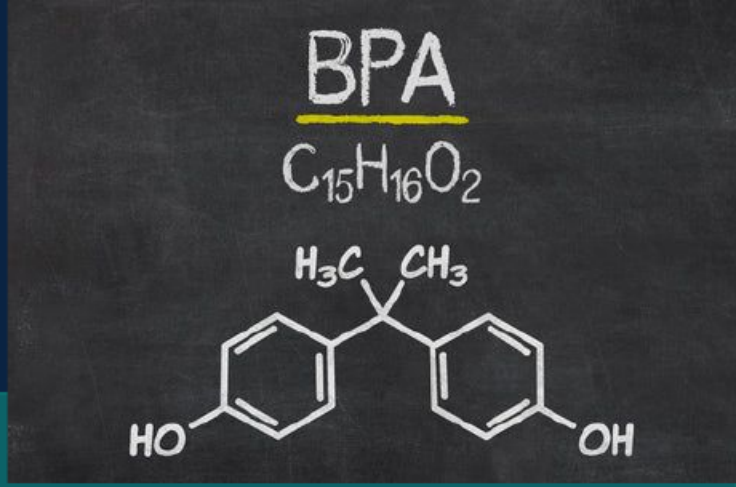


Human Health Risk Assessment of Bisphenol A (BPA) Exposure in North American Adults via Urinary Biomonitoring

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Non-Cancer Human Health Risk Assessment · Hazard Quotient Approach

Problem Formulation

Suspected endocrine-disrupting chemical with plausible pathways for human exposure
Adverse effects observed in animal studies at low concentrations

Effects of Concern

- Mimics estrogen by binding to ERα/ERβ receptors
- Potential effects: reproductive toxicity, thyroid disruption, metabolic changes

Pathways of Exposure

- Primary: Oral ingestion (polycarbonate containers, epoxy cans)
- Secondary: Dermal contact (thermal paper)
- Negligible: Inhalation (low vapor pressure)

Severity of Effects

- Moderate-to-serious disruption at low doses
- EFSA (2023) recent revision suggests the threshold for harm maybe lower than thought

Biological Plausibility

- BPA similar to diethylstilboestrol (DES)
- Can accumulate in tissues; detected in >90% of adults' urine

Objective & Rationale

OBJECTIVE

Estimate daily BPA intake in North American adults using urinary biomonitoring data from NHANES (U.S.) and CHMS (Canada), convert measurements to estimated daily intake (EDI), and compare against regulatory health benchmarks using the Hazard Quotient approach.

01 Ubiquitous Exposure

- BPA is found in polycarbonate plastics and epoxy resins
- Used in food packaging and containers
- Routine oral exposure is plausible for a large proportion of the population

02 Measurable Biomarker

- BPA is metabolized and excreted in urine within ~24 hours
- NHANES and CHMS data provide population-level estimates of internal exposure
- Supports a biologically meaningful dose estimate

03 Regulatory Significance

- EFSA (2023) reduced its TDI by 20,000-fold to 0.2 ng/kg bw/day based on immune/metabolic endpoints
- Raises questions about whether current North American exposures exceed updated health benchmarks

Stressor Information: Bisphenol A (BPA)

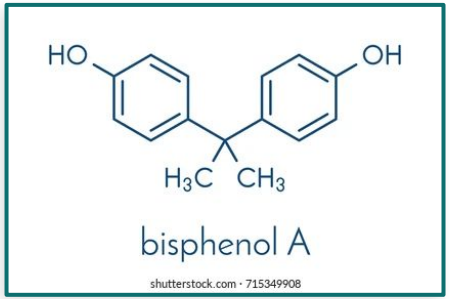
Chemical Identity

Common name:	Bisphenol A (BPA)
Systematic name:	4,4'-(propane-2,2-diyl) diphenol
Molecular formula:	$C_{15}H_{16}O_2$
Molecular weight:	228.29 g/mol
CAS Number:	80-05-7
Physical state:	White crystalline solid (at 25°C)
Primary uses:	Polycarbonate plastics, epoxy resin can linings, thermal paper

Key Physico-Chemical Properties — Relevance to This Risk Assessment

Property	Value	Why it matters for this RA
Log Kow	3.64 ± 0.32	Moderate lipophilicity → potential accumulation in tissues; supports dietary oral exposure pathway
Aqueous Solubility	120–257 mg/L @ 25°C	Readily dissolves in beverages and aqueous food matrices → oral ingestion plausible
Vapour Pressure	5×10^{-6} Pa @ 25°C	Extremely low → inhalation route is negligible; justifies oral-only focus in this assessment
Henry's Law Const.	1.0×10^{-6} Pa·m³/mol	Low volatility from water → BPA remains in aqueous/food matrices, not in air

BPA Structural Formula

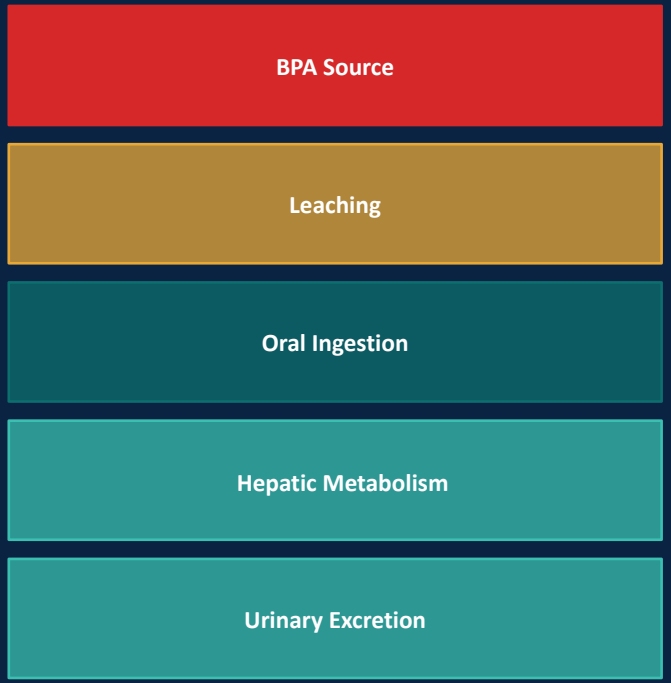


Receptor Information: North American Adults

Receptor Profile

Receptor:	North American adult population (≥18 years)
Datasets:	U.S. adults — CDC NHANES Canadian adults — Health Canada CHMS
Body weight (default):	70 kg (Health Canada / USEPA adult default)
Exposure route assessed:	Oral ingestion only (primary pathway for BPA)
Detection prevalence:	>90% of adults show measurable urinary BPA (NHANES / CHMS population surveys)
Biological mechanism:	BPA rapidly glucuronidated in liver and excreted in urine — urinary concentration reflects recent dietary exposure
Excluded subgroups:	Infants, children, pregnant women (higher vulnerability — assessed separately in literature)

Oral Exposure Pathway



Methodology: Exposure Profile

DATA SOURCE

CDC NHANES (National Health and Nutrition Examination Survey) — urinary BPA concentrations (µg/L) in U.S. adults
CHMS (Health Canada) used for Canadian context comparison

NHANES URINARY BPA DATA (µg/L)

Exposure Scenario	Urinary BPA (µg/L)	Represents
Median (50th percentile)	~1.1 µg/L	Typical adult in population
95th percentile	~7.0 µg/L	High-end exposure scenario

EDI CALCULATION

Estimated Daily Intake (EDI) = [Urinary BPA (µg/L) × Daily Urine Volume (L/day)] ÷ Body Weight (kg)

Median: 1.1 µg/L × 1.75 L/day ÷ 70 kg = 0.0275 µg/kg bw/day ≈ 0.028 µg/kg bw/day
95th pct: 7.0 µg/L × 1.75 L/day ÷ 70 kg = 0.175 µg/kg bw/day

Methodology: Effects Profile & Risk Characterization

TOXICITY BENCHMARKS

Agency / Source	Benchmark	Value	Basis Endpoint	Year
EFSA (Europe)	**TDI**	**0.2 ng/kg bw/day**	**Immune & metabolic effects ($BMDL_{10}$ from animal studies)**	**2023**
Health Canada	TDI	25 µg/kg bw/day	Reproductive/developmental NOAEL (rat)	2012
USEPA	RfD (oral)	50 µg/kg bw/day	Reproductive toxicity (rat NOAEL)	Ongoing
EFSA (prior)	TDI	4 µg/kg bw/day	Reproductive/developmental NOAEL	2015

RISK CHARACTERIZATION METHOD

Hazard Quotient (HQ) = EDI ÷ TDI

HQ < 1 → Exposure below health-based threshold; risk may be acceptable
HQ > 1 → Exposure exceeds guideline; potential health concern warranting further review

Results: Risk Characterization

ESTIMATED DAILY INTAKE (EDI)

50th Percentile (Typical Adult)

~0.028 µg/kg bw/day

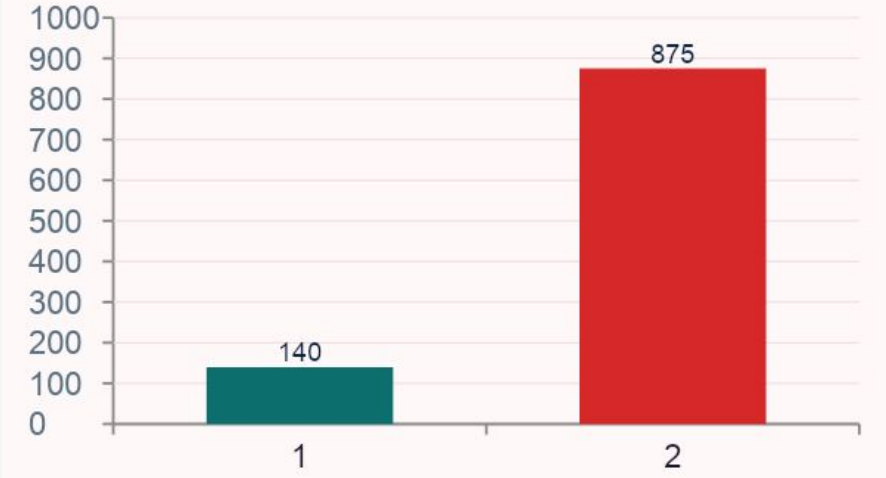
1.1 µg/L × 1.75 L/day ÷ 70 kg = 0.0275 µg/kg bw/day

95th Percentile (High-End Exposure Scenario)

~0.175 µg/kg bw/day

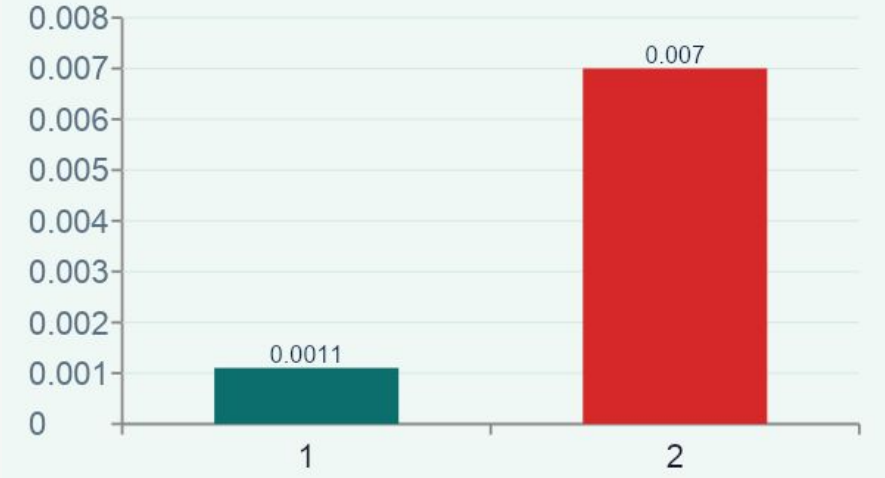
7.0 µg/L × 1.75 L/day ÷ 70 kg = 0.175 µg/kg bw/day

HQ — EFSA 2023 TDI (0.2 ng/kg/day)



Both HQ >> 1 → Potential concern under EFSA 2023

HQ — Health Canada TDI (25 µg/kg/day)



Both HQ << 1 → No concern under Health Canada TDI

Discussion

Objective Achieved

- Estimated daily BPA intake in North American adults
- Used NHANES biomonitoring data
- Compared against regulatory benchmarks using Hazard Quotient approach
- Evaluated at median and 95th percentile levels

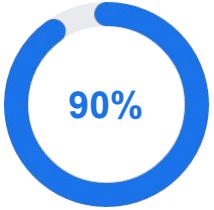
How to Improve This Assessment

- Monte Carlo simulation to propagate uncertainty across EDI distributions
- Stratify by age, sex, and reproductive status
- Include dermal route (thermal paper handling)
- Use biomonitoring equivalents (BEs) to cross-validate EDI estimates

Benchmark Choice Substantially Affects Interpretation

- Benchmark selection affects risk interpretation:
- Health Canada TDI (25 µg/kg/day) suggests no concern
- EFSA 2023 TDI (0.2 ng/kg/day) suggests HQ > 100 at median exposure
- Reflects evolving science on low-dose endocrine effects

Conclusion



BPA detectable in N. American adults

Median Intake

0.028 µg/kg/day

Risk Comparison

EFSA 2023: Potential concern

Health Canada: No concern

Key Assumptions & Limitations

- 100% urinary excretion assumed
- Fixed urine volume (1.75 L/day)
- Dermal and dust routes excluded

Assessment Improvements

- Monte Carlo uncertainty simulation
- Stratify by age and sex
- Include dermal route (thermal paper)

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